Prerana Acharyya

Bachelor of Engineering
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Summary

Full-Stack Developer and Jadavpur University student with experience in React.js, Node.js, RESTful APIs, AI integrations, and machine learning. Strong problem-solving skills with a Codeforces Specialist rating and 160+ LeetCode problems solved. Experienced in building web applications, AI-driven workflows, and ML models.

Technical Skills

- **Languages:** JavaScript (ES6+), TypeScript, Java, Python, C, SQL, HTML5, CSS3, JSON
- **Machine Learning & Data Analysis:** Pandas, NumPy, Scikit-learn, CatBoost, XGBoost, Matplotlib, Google Colab
- **Frontend:** React.js, Tailwind CSS, React Router, Responsive Web Design (CSS Grid/Flexbox), DOM Manipulation
- **APIs & Integration:** RESTful APIs, Axios, RapidAPI, Postman
- **Tools & Platforms:** Git/GitHub, Vercel, Render, Vite, Node.js, VS Code, IntelliJ IDEA
- **Core CS:** Data Structures, Algorithms, Object-Oriented Programming (OOP), DBMS

Programming Profiles & Achievements

- **Codeforces:** **Specialist (Rating: 1548)**; Rank: Top 15% global.
- **LeetCode:** **Rating: 1714 (Top 12.2%)**; Solved 160+ problems.
- **IIT Bombay (Azeotropy)|OPTIMIZER(2025):** Finalist (Rank 6th).

Experience

- **CATI AI [ ]** *July 2026 – October 2026*
  SDE Intern Remote
  – Developed features for an **AI voice-agent platform**, working across AI agent workflows, API integrations, function calling, frontend interfaces, and testing.
  – Built **Calendly REST API integration** using Node.js and Axios, enabling AI agents to list event types, check availability, book appointments, send confirmations, and handle rescheduling.
  – Integrated **HubSpot CRM** using OAuth and REST APIs, with JSON-based tool definitions and function-calling workflows for contact, company, deal, note, activity, CRM search, and appointment-related actions during AI voice calls.
  – Built reusable **goal-based AI agent templates** for sales, lead qualification, appointment booking, customer support, and feedback, with configurable business information, knowledge, voice, and language; also developed **healthcare AI prompt templates** for hospitals, clinics, and diagnostic centers.
  – Developed the **Test Agent interface** and testing workflows, including API test scripts, endpoint validation, booking-flow testing, authentication/connection testing, error handling, and multilingual voice-agent testing.

Personal Projects

- **Machine Learning-Based Prediction of Biomass Pretreatment** *January 2026 – April 2026*
  Tech Stack: Python | Pandas | NumPy | Scikit-learn | CatBoost | XGBoost | Matplotlib | Google Colab [Live Demo]
  – Developed a machine learning pipeline to predict **lignin removal yield** using biomass, process, solvent, and molecular fingerprint features.
  – Preprocessed and combined literature-based datasets and performed feature engineering using Pandas and NumPy.
  – Trained and compared **11 regression models**, including CatBoost, Random Forest, XGBoost, SVR, and ANN.
  – Evaluated models using R^2 and RMSE, with **CatBoost achieving $R^2 = 0.830$ and RMSE = 0.073**.

- **Football Analytics Dashboard** *April 2025*
  Tech Stack: HTML5 | CSS3 | JavaScript (ES6+) | Chart.js | Node.js | Express.js | REST API [| Live Demo]
  – Developed a full-stack dashboard to display live football league standings and team statistics across multiple leagues.
  – Built a Node.js and Express.js backend to fetch data from the Football-Data.org REST API and serve it to the frontend.
  – Implemented team search, sorting, and interactive Chart.js visualizations for comparing team performance.
  – Secured API credentials using environment variables and a backend proxy, with the frontend and backend deployed separately on Netlify and Render.

Education

Course Name	Institute Name	University Name	Year of Passing	Score Details (%/CGPA)	Course Type
B.E. in Chemical Engineering	Jadavpur University	Jadavpur University	2027 (Expected)	7.71/10 SGPA (Upto 6th Sem)	Full-time
B.S. in Data Science and Programming	IIT MADRAS	IIT MADRAS	2024	9.38/10 CGPA (Upto Foundation Level)	Online
Higher Secondary Education	Pathfinder Higher Secondary Public School	–	2023	85.6%	Full-time
Secondary Education	Gyandeep Vidyapith	–	2021	98.86%	Full-time